CSE 4345: Big Data Analytics

Week 8, Part 1 — Data Integration at Scale: Pipelines, Data Lakes & Modern Architectures

Department of Computer Science & Engineering
Premier University, Chittagong

Semester 2026

Course & Lecture Information

Lecture Identity

Course:	CSE 4345
Week / Part:	8 of 11 | Part 1 of 2
CLO:	CLO3, CLO4
PLO:	PLO(e), PLO(f)

CLO3 & CLO4

CLO3: *Explain big data techniques and tools to store, manage, and analyse large datasets*

CLO4: *Solve big data problems and build a project applying big data tools and frameworks*

Course Progression

Week 6: Spark, RDDs, in-memory computing

Part 1 Covers

- From ETL to pipelines: the shift in integration thinking
- Data integration tool landscape (Sqoop, Flume, Kafka, NiFi)
- Apache Sqoop: bulk RDBMS $\leftrightarrow$ HDFS transfer
- Apache Flume: log ingestion pipelines
- Apache Kafka: append-only logs, topics, partitions, offsets
- Kafka producers, consumers, consumer groups, and retention
- Lambda architecture (Marz): Batch + Speed + Serving layers
- Kappa architecture (Kreps): stream-only

Lecture Agenda

1. From ETL to Pipelines: A Shift in Architecture
2. Batch Integration: Sqoop and Flume
3. Apache Kafka: The Append-Only Event Log
4. Lambda Architecture
5. Kappa Architecture
6. Data Lakes and the Data Swamp Risk
7. Ivy League Alignment
8. Student Assessment Pool

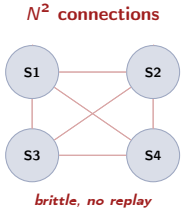
From ETL to Pipelines: A Shift in Architecture

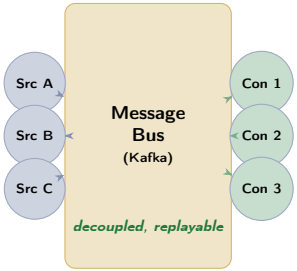
Why Pipelines Replace Point-to-Point ETL

The Old Model: Point-to-Point ETL

Traditional enterprises connected each pair of systems with a direct ETL job. With N systems, this creates $O(N^2)$ point-to-point connections. At 50 systems: potentially 2,450 separate ETL jobs to maintain.

- Brittle: every schema change in one system breaks its ETL jobs
- No replay: if a downstream consumer was offline, data is lost
- No real time: all jobs are nightly batch; latency $\geq$ 24 hours
- No decoupling: source and consumer are tightly bound





The New Model: Event-Driven Pipeline

Data Integration Tool Landscape

Tool	Direction	Type	Primary Use Case	Throughput
Apache Sqoop	RDBMS ↔ HDFS	Batch	Nightly bulk transfer of Oracle/MySQL tables to Hadoop	GB – TB/hour
Apache Flume	Logs → HDFS	Batch / micro-batch	Web server log ingestion; aggregating multi-source logs	MB – GB/hour
Apache Kafka	Any → Any	Real-time stream	Centralised event bus; fraud events, clickstream, IoT	GB – TB/hour
Apache NiFi	Any → Any	Visual pipeline	No-code data routing; IoT edge to cloud; file transfer	MB – GB/hour
Spark Streaming	Kafka → HDFS/DB	Stream processing	Stateful aggregation on streams; enrichment;	GB/hour

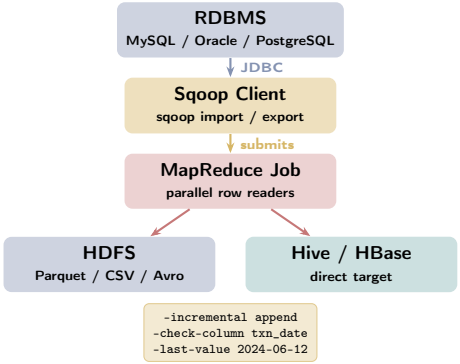
Batch Integration: Sqoop and Flume

Apache Sqoop: Bulk RDBMS ↔ HDFS Transfer

What Sqoop Does

Sqoop (**SQL-to-Hadoop**) is a command-line tool that transfers data in bulk between a **relational database** (MySQL, Oracle, PostgreSQL, SQL Server) and **HDFS** (or Hive, HBase).

- **Import**: copies a table or query result from RDBMS → HDFS as CSV, Avro, Parquet, or Sequence File
- **Export**: copies data from HDFS → RDBMS (e.g., after Spark processing, push results back to MySQL)
- Internally launches a **MapReduce job** where each Map task reads a subset of rows (partitioned by primary key range) in parallel
- Supports **incremental import**: only rows newer

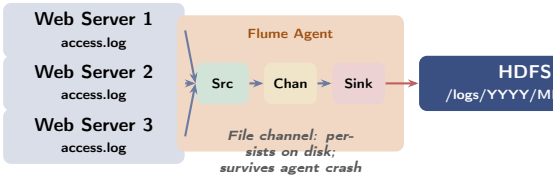


Apache Flume: Log Ingestion Pipelines

Flume Architecture: Source → Channel → Sink

Flume moves **semi-structured log data** (web server logs, application logs, IoT sensor readings) into HDFS for archival and batch analysis. Every Flume agent has three components:

- **Source**: reads incoming data. Types: Avro source (from another agent), Syslog source, Exec source (tail a log file), HTTP source
- **Channel**: buffers events between Source and Sink. Memory channel (fast, no persistence) or **File channel** (persistent, survives agent restart)
- **Sink**: writes data to the destination. HDFS sink (writes Avro/text files), Kafka sink (forward to Kafka), HBase sink



Flume vs. Kafka

Flume is optimised for **log collection to HDFS**. Kafka is a **general-purpose event bus** with stronger durability and replay semantics. In modern stacks, Flume is often replaced by Kafka producers writing directly from applications.

Apache Kafka: The Append-Only Event Log

Apache Kafka: The Core Insight

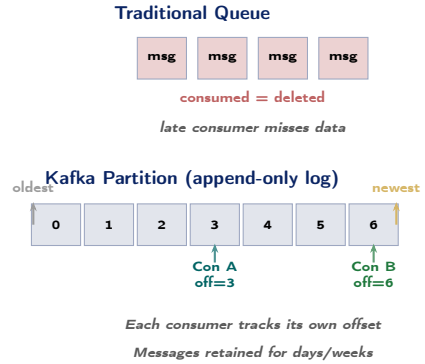
Origin: LinkedIn, 2011

LinkedIn needed to move **activity events** (profile views, connection requests, job clicks, feed impressions) from dozens of producer services to dozens of consumer services (recommendation engines, analytics, monitoring). Point-to-point messaging created an N^2 web of dependencies.

Jay Kreps, Neha Narkhede, and Jun Rao designed Kafka around one insight:

The Founding Insight

*"A log is the most primitive form of storage. If you model your data infrastructure as a distributed, replicated, append-only log, you get **durability**, **replayability**, and **decoupling** for free."*

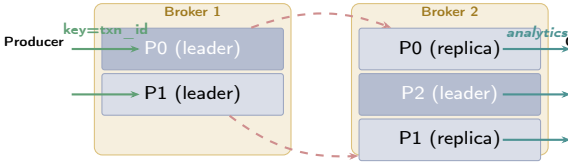


Kafka Architecture: Topics, Partitions, Brokers

Key Components

- **Topic**: a named, durable, append-only log for a category of events (e.g., `bkash-transactions`, `pathao-gps-pings`)
- **Partition**: each topic is split into P ordered partitions for parallelism. All messages with the same **key** go to the same partition (preserving key-level ordering)
- **Offset**: integer index of a message within a partition. Each consumer independently tracks its own read offset
- **Broker**: a Kafka server that stores partitions on disk. A cluster has B brokers; each partition has one **leader** and $R - 1$ **follower replicas** (default $R = 3$)
- **Producer**: publishes messages to a topic; selects

Topic: `bkash-txns` (3 partitions, RF=2)



Kafka: Throughput, Retention, and Replay

How Kafka Achieves High Throughput

- **Sequential disk writes**: Kafka never updates messages in place. Append-only writes are sequential — modern HDDs achieve 600 MB/s sequential vs. <1 MB/s random. Kafka exploits this fully.
- **Zero-copy transfer**: Kafka uses the OS `sendfile()` system call to transfer data from disk to network socket without copying through user-space memory. Eliminates one full memory copy per message.
- **Batching**: producers and consumers batch multiple messages in one network round-trip. Configurable via `batch.size` and `linger.ms`.
- **Partition parallelism**: P partitions enable P producers and up to P consumers in a group to

Kafka at bKash and Pathao

bKash uses Kafka topics `wallet-credit` and `wallet-debit` to stream every transaction event to three consumers simultaneously:

1. Fraud detection (Spark Streaming, <200 ms latency)
2. Balance update (RDBMS writer)
3. Audit log writer (HDFS, daily reconciliation)

Pathao publishes GPS pings every 3 seconds from active drivers to a Kafka topic `driver-location` (10 partitions, keyed by `driver_id`), consumed by the routing engine and the dispatch service independently.

Retention and Replay

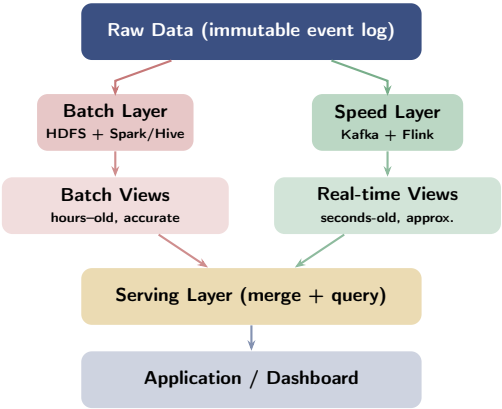
Lambda Architecture

Lambda Architecture (Nathan Marz, 2011)

Design Goal

Serve **both** accurate historical batch views and **low-latency** real-time views from the same data, with fault tolerance and the ability to recompute from scratch. **Three layers:**

1. **Batch Layer**: stores the complete, immutable master dataset (HDFS); periodically recomputes **batch views** (e.g., Spark/Hive jobs running every hour or nightly). Output is accurate but **high-latency** (1 hour to 24 hours stale).
2. **Speed Layer**: processes new data arriving since the last batch job using a stream processor (Kafka + Spark Streaming / Flink). Produces **real-time views** covering the most recent window. Output is **approximate but low-latency** ($\lesssim$seconds).



Lambda Architecture: Strengths and Weaknesses

Strengths of Lambda

- **Accuracy**: the batch layer always recomputes from raw data, so historical views are correct regardless of bugs or schema changes in earlier runs
- **Low latency**: the speed layer delivers results within seconds for the most recent data window
- **Fault tolerance**: the immutable master dataset on HDFS is the single source of truth; any derived view can be recomputed from it
- **Separation of concerns**: batch and speed components can be built, deployed, and scaled independently

The Critical Weakness: Dual Code Paths

Lambda requires implementing **the same business logic twice**:

- Once in the batch layer (Spark/Hive jobs)
- Once in the speed layer (Flink/Spark Streaming)

When business logic changes, **both implementations must be updated simultaneously and kept in sync**. Teams report that 30–40% of engineering effort goes into keeping the two layers consistent — a significant maintenance burden. **Additional complexity:**

- The serving layer must merge two query results at runtime (complex query logic)

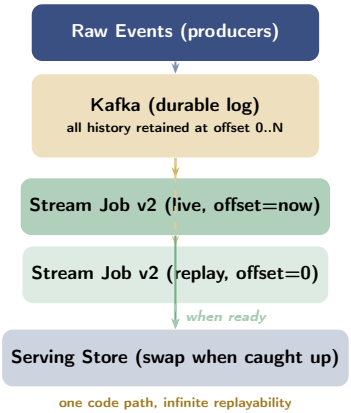
Kappa Architecture

Kappa Architecture (Jay Kreps, 2014)

The Simplifying Insight

Jay Kreps (co-creator of Kafka) argued that the batch layer in Lambda is unnecessary if your stream processing system is **powerful enough to reprocess historical data** at the same speed as it processes live data. **Kappa's two components:**

1. **Log storage** (Kafka with long retention or Apache Iceberg on S3): stores the complete immutable event history. Replacing the HDFS batch layer.
2. **Stream processing engine** (Flink, Spark Structured Streaming): the **single code path** that handles both live events and historical replay. To reprocess history, start a new consumer at offset 0 and let it catch up.



Lambda vs. Kappa: Full Comparison

Property	Lambda Architecture	Kappa Architecture
Code paths	Two (batch + stream) — must stay in sync	One (stream only)
Historical reprocessing	Re-run batch job on HDFS	Replay Kafka log from offset 0
Latency	Low (speed layer) + High (batch layer)	Low throughout
Complexity	High (two systems, merging layer)	Lower (single pipeline)
Batch throughput	Very high (Spark/Hive on full dataset)	Limited by stream engine throughput
Long-term storage	HDFS (immutable master dataset)	Kafka (bounded) + external store (S3/Iceberg)
When to use	Business logic too complex for	—

Data Lakes and the Data Swamp Risk

Data Lake Architecture

What Is a Data Lake?

A data lake is a centralised repository that stores **raw data at any scale, in any format** (structured, semi-structured, unstructured) without requiring upfront schema definition. Data is ingested cheaply and quickly; transformation happens on demand.

Four-layer architecture:

1. **Ingestion Layer**: Sqoop, Flume, Kafka, NiFi, REST APIs — pulls data from all sources and writes raw files to storage
2. **Storage Layer**: HDFS or S3 / Azure ADLS — object storage organised in zones (raw → processed → curated)
3. **Processing Layer**: Spark, Hive, Presto — transforms and queries data on demand

The Data Swamp Problem

A data lake without governance rapidly becomes a **data swamp**: raw files accumulate without metadata, owners, or quality guarantees.

Symptoms:

- Analysts cannot find relevant datasets (no catalogue)
- No one knows which files are current vs. stale
- Multiple copies of the same data with different transformations exist with no version control
- PII (personal data) is mixed with non-sensitive data, violating privacy regulations

Prevention: enforce a data catalogue at ingest time; assign owners to every dataset; implement

Ivy League Alignment

Pipelines & Architectures in Top University Curricula

MIT 6.830 — Database Systems

MIT frames the Lambda and Kappa architectures as answers to a classical database question: **how do you serve both OLAP (batch analytics) and OLTP (real-time transactions) from the same data?**

- Lambda = a distributed version of the classical data warehouse + OLTP dual architecture
- Kappa = a stream-first architecture that eliminates the data warehouse entirely, replacing it with a replayable log
- MIT students implement a mini Kafka (topics, partitions, producers, consumers) from scratch in Go as a course project — reinforcing that the append-only log is the fundamental primitive, not a specific product

MIT exam question: *"Why does Kafka use sequential disk writes rather than a B-tree index? Quantify the throughput difference on spinning disk."*

Stanford CS246 — Mining Massive Datasets

Stanford covers Kafka as the canonical example of **durable, replayable distributed log storage** — the same abstraction as Google's distributed log (used internally in Google's Spanner and BigTable replication). CS246 uses Kafka + Spark Structured Streaming as the lab environment for real-time PageRank computation on a Twitter follow graph.

CMU 15-721 — Advanced DB Systems (Pavlo)

CMU compares Kafka's log-structured storage to LSM-trees (Log-Structured Merge trees) used in RocksDB, Cassandra, and HBase — all

Student Assessment Pool

Instructions

Select the single best answer. Correct answers **bolded** for instructor reference.

Q1. Which tool is specifically designed to transfer data in bulk between an RDBMS (e.g., MySQL) and HDFS?

a) Apache Flume
b) Apache Kafka
c) **Apache Sqoop**
d) Apache NiFi

Q2. Apache Kafka's data model is based on which fundamental storage abstraction?

a) B-tree indexes for fast point lookups
b) Column-family storage (like HBase)
c) **Append-only immutable distributed logs (partitioned topics)**
d) Document storage with secondary indexes

Instructor Notes

Q1: Flume (a) is for logs to HDFS; Kafka (b) is for real-time streaming; NiFi (d) is a visual pipeline

Q3. In the Lambda architecture, which layer handles **real-time, low-latency** query views?

(a) Batch Layer (b) Serving Layer (c) **Speed Layer** (d) Ingestion Layer

Q4. The Kappa architecture differs from the Lambda architecture primarily by:

(a) Adding a third processing layer for machine learning
(b) Using only relational databases for both batch and stream storage
(c) **Eliminating the batch layer and using Kafka log replay for all reprocessing**
(d) Replacing Hadoop with Spark in the batch layer while keeping the speed layer

Instructor Notes

Q3: The Serving Layer (b) is not a processing layer — it merges precomputed views and serves queries. The Batch Layer (a) is high-latency. The Speed Layer (c) is the real-time component. **Q4:** Option (d) is a common misconception — Kappa does not just replace Hadoop; it eliminates the entire batch layer and relies on a single stream processing code path with log replay for historical reprocessing.

Assessment—True / False

CLO3

Instructions

State True or False and provide a **one-sentence justification** for full marks.

Statement	Answer
1. Apache Sqoop is designed for real-time streaming ingestion from transactional systems.	**False**
2. Kafka retains messages for a configurable period, allowing consumers to replay events from any past offset.	**True**
3. The Lambda architecture requires maintaining two separate code paths implementing the same business logic.	**True**
4. Apache Flume is the preferred tool for collecting and aggregating server log data into HDFS.	**True**

Assessment—Descriptive Questions *CLO3*

Instructions

Answer each question in **100–150 words**. Include a diagram where indicated.

D1. Explain the **Lambda architecture** with a labelled diagram. What are its three layers? What is its primary engineering disadvantage compared to the Kappa architecture?

D2. Describe how Apache Kafka achieves **high throughput and fault tolerance**. Explain the roles of topics, partitions, offsets, brokers, and replication factor in the Kafka data model.

D3. What is a **data lake**? How does it differ from a data warehouse in terms of schema enforcement, storage format, and typical use cases? What is a "data swamp" and how is it prevented?

Assessment—Analytical Questions

CLO4

Instructions

Answer in **200–300 words** with step-by-step reasoning and tool justifications.

A1. Pipeline Design for a Bank:

Design a data integration pipeline for a Bangladeshi commercial bank that must:

- (a) Synchronise 10 million customer records nightly from Oracle to Hadoop for analytics
- (b) Detect fraudulent card-swipe events in real time with a decision within 200 ms
- (c) Power a risk management dashboard with data no older than 5 minutes

Specify which tool (Sqoop, Flume, Kafka, Spark Streaming) you would use at each stage. Justify each choice with reference to latency, throughput, and fault-tolerance requirements. Which architecture (Lambda or Kappa) fits best, and why?

A2. Lambda to Kappa Migration:

A team using Lambda architecture complains that maintaining two code paths (batch + stream) doubles development cost. Their CTO suggests migrating to Kappa. Analyse: (a) what must be true about the stream processing engine for Kappa to be viable; (b) what are the risks of migrating; (c) under what conditions is Kappa **not suitable** (give at least two concrete scenarios from the Bangladesh financial or healthcare sector)?

Textbook & Reading References

Primary Textbooks for Week 8

- **[TW]** Tom White. *Hadoop: The Definitive Guide*, O'Reilly, 4th ed., 2015. **Ch. 14, 15**
  - Ch. 14: Apache Flume — architecture, sources, channels, sinks, fan-in/fan-out
  - Ch. 15: Apache Sqoop — import/export, incremental, partitioning
- **[SA]** Acharya & Chellappan. *Big Data and Analytics*, Wiley, 2015. **Ch. 8**
  - Ch. 8: Data integration tools and architectures overview
- **[TE]** Erl, Khattak & Buhler. *Big Data Fundamentals*, Prentice Hall, 2016. **Ch. 10**
  - Ch. 10: Big Data integration patterns and enterprise architecture considerations

Graduate Reference and Seminal Paper (covered in Week 8, Part 2)

- Kreps, J., Narkhede, N., & Rao, J. (2011). **Kafka: A Distributed Messaging System for Log Aggregation.** LinkedIn Engineering. [PRIMARY SEMINAL PAPER]
- Kleppmann, M. (2017). *Designing Data-Intensive Applications.* O'Reilly. Ch. 3 (storage engines), Ch. 10 (batch processing), Ch. 11 (stream processing). [Graduate-level systems reference at MIT, CMU, Stanford]

Questions & Discussion

Week 8, Part 1 | CSE 4345 Big Data Analytics

"I believe that a unified log is the central data structure that can solve the coherency problem in a distributed, heterogeneous data system — a structure so simple that we take it for granted, and so fundamental that we underestimate it."

— Jay Kreps, *The Log*, LinkedIn Engineering, 2013

Next Lecture: Week 8, Part 2

Seminal Paper: Kreps, Narkhede & Rao (2011) — *"Kafka: A Distributed Messaging System for Log Aggregation"*

Case Study: **Netflix Keystone Real-time Stream Processing Platform** — 500 billion events per day, Kafka + Flink + S3 + Apache Iceberg

Reading before Part 2: [TW] Ch. 14–15 · Kreps et al. (2011) Kafka paper (see references)